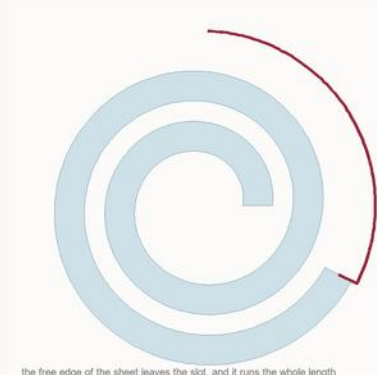
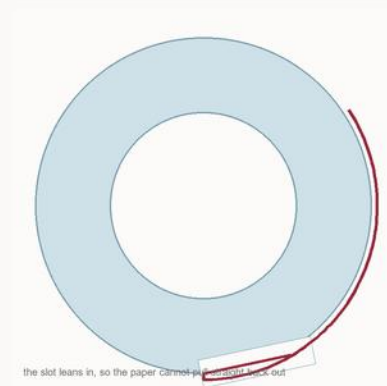


Glass tip — two ways

Same job, opposite answers. One is a tube with a slot in it. The other is a rolled sheet, and the roll is the slot. Sections at right, with the paper in.



Slotted tube

A slot the whole length, out both ends, raked seventy-eight degrees off the radius so it undercuts. Three millimetres of paper goes in. Thick wall, and a perforated screen across the bore.

Overall	19 mm
Outside	ø 9 mm
Bore	ø 5.0, 2.0 wall
Screen	seven ø 1.25 holes
Slot	0.75 wide, 3 mm of grip
Mass	≈ 1.4 g

Rolled sheet

No tube at all - a thin sheet rolled into a 9 mm cylinder, and the stock cut into lengths. The gap between the wraps is the slot, it runs the whole piece, and the free edge is where the paper goes in.

Overall	19 mm
Outside	ø 9 mm
Sheet	0.8 thick
Gap	0.55 between wraps
Wraps	about two and a half
Mass	≈ 1.15 g

Paper on, both ways

Three millimetres of the paper goes into the slot, along its whole length. Then it rolls.



Slotted tube

paper in the slot



Slotted tube

paper on



Rolled sheet

paper into the outer ring

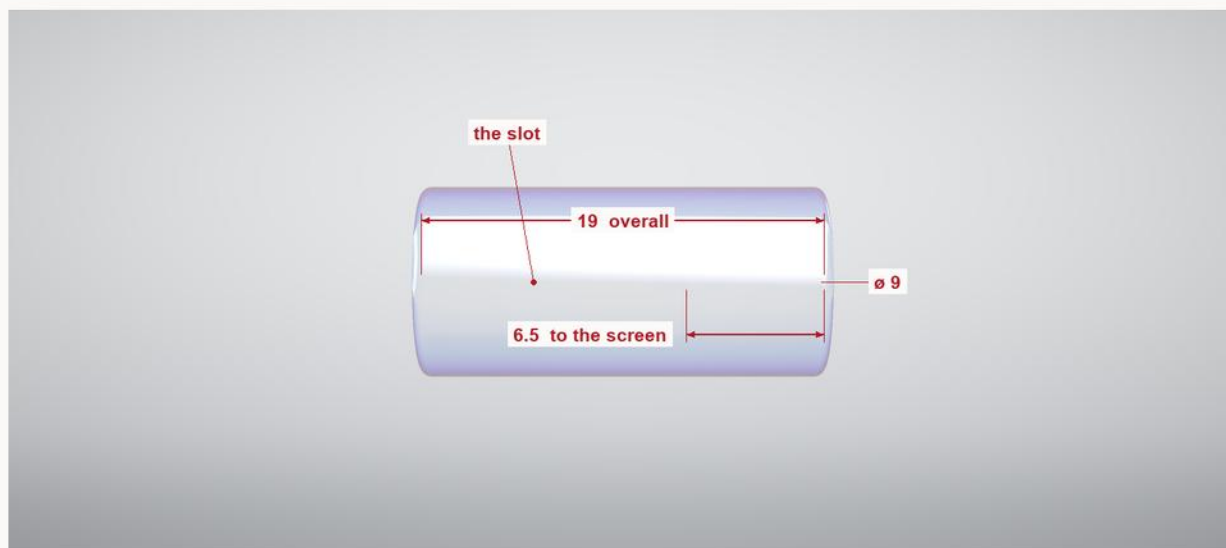


Rolled sheet

paper on

Glass tip

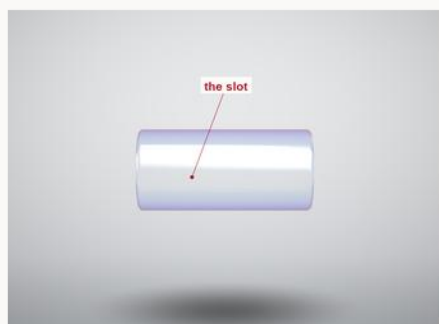
19 mm of clear 9 mm tube. Perforated screen across the bore, and a straight slot cut at a long oblique to start the roll.



Down the bore
seven holes, one centre

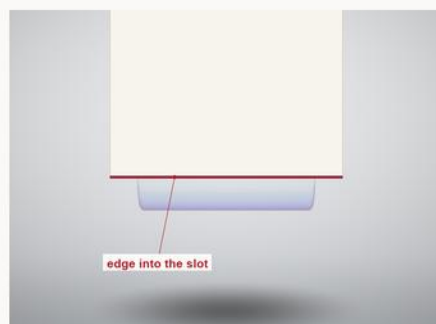
Overall length
19 mm
Outside diameter
9 mm
Bore / wall
6.4 / 1.3 mm
Screen
1.5 thick, 6.5 in
Screen holes
seven ø 1.25
Slot
0.75 wide, 3 mm of grip
Slot rake
68° off the axis
Glass
clear boro 3.3, ≈ 1.4 g

ROLLING



One

The slot runs the length of the tube, cut in on a lean.



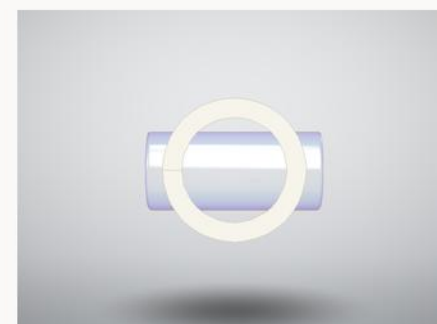
Two

Slip the leading edge of the paper into it. The slot holds the paper on its own.



Three

Roll the tip. The paper winds on, and the rake starts the wrap slightly off-square - the bias a cone wants.



Four

Wound on and filled. The screen keeps it from pulling through, and the tip washes and goes again.